Windows Security Log Analysis Report

1. Project Overview

Project Title:	**Windows Security Event Log Analysis**
Analyst:	Vaibhavi Singh
Date of Analysis:	12 October 2026
Tools Used:	Windows Event Viewer, Google Sheets

Objective:

The goal of this project was to analyze Windows Security Event Logs to identify normal user activity, detect suspicious behavior, and understand how security incidents are recorded at the operating system level. This project focuses on three specific Event IDs: **4624** (Successful Logon), **4625** (Failed Logon), and **4798** (User Group Enumeration).

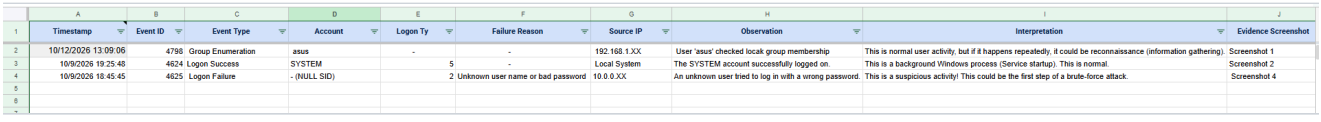
2. Methodology

The analysis was performed using the following steps:

1. **Data Collection:** Windows Event Viewer was opened, and the "Security" log was accessed under Windows Logs.
2. **Filtering:** The logs were filtered by specific Event IDs (4624, 4625, and 4798) to focus on security-relevant activities.
3. **Data Extraction:** Relevant fields such as Timestamp, Event ID, Account Name, Logon Type, Failure Reason, and Source IP were extracted and organized into a structured Google Sheet.
4. **Interpretation:** Each event was analyzed in context (time, frequency, logon type) to determine whether it was a normal activity or a potential security threat.
5. **Privacy Protection:** All sensitive information, including IP addresses and computer names, was masked (e.g., `192.168.1.XX`) to protect the privacy of the local environment.

3. Complete Data Sheet (Google Sheets)

The following screenshot shows the complete data extracted from Windows Event Viewer and organized in Google Sheets for analysis:



	A	B	C	D	E	F	G	H	I	J
1	Timestamp	Event ID	Event Type	Account	Logon Ty	Failure Reason	Source IP	Observation	Interpretation	Evidence Screenshot
2	10/12/2026 13:09:06	4798	Group Enumeration	asus	-	-	192.168.1.XX	User 'asus' checked locak group membership	This is normal user activity, but if it happens repeatedly, it could be reconnaissance (information gathering).	Screenshot 1
3	10/9/2026 19:25:48	4624	Logon Success	SYSTEM	5	-	Local System	The SYSTEM account successfully logged on.	This is a background Windows process (Service startup). This is normal.	Screenshot 2
4	10/9/2026 18:45:45	4625	Logon Failure	- (NULL SID)	2	Unknown user name or bad password	10.0.0.XX	An unknown user tried to log in with a wrong password.	This is a suspicious activity! This could be the first step of a brute-force attack.	Screenshot 4
5										
6										

Full Data Sheet Link: [Click here to view the complete Google Sheet]

Direct URL: https://docs.google.com/spreadsheets/d/1CGy4Wghq2sfYyBpzHdPe5Yel5BEDveXoMsBwzxQsuws/edit?usp=sharing

4. Event ID Analysis (Based on the Sheet)

Event ID 4624: Successful Logon

Description: This event indicates that an account successfully logged on to the system.

Observation:

- **Timestamp:** 10/9/2026 7:25:48 PM
- **Account Name:** SYSTEM
- **Logon Type:** 5 (Service)
- **Source IP:** Local System

Interpretation:

This event represents a background Windows service starting up. The account "SYSTEM" is a built-in Windows account used for internal processes. Logon Type 5 indicates that a service was started, which is a normal and expected activity.

Verdict: Normal Activity.

Event ID 4625: Failed Logon

Description: This event indicates that an account failed to log on, typically due to an incorrect username or password.

Observation:

- **Total Events:** 7
- **Timestamp Range:** 10/6/2026 to 10/12/2026
- **Account Name:** asus (in some events) and NULL SID (in others)
- **Logon Type:** 2 (Interactive/Local)
- **Failure Reason:** Unknown user name or bad password

Interpretation:

There are 7 failed logon attempts recorded. Key observations:

1. **Time Frequency:** The attempts occurred on different days (Oct 6, 9, 10, 11, 12), not in a single burst. A brute-force attack typically involves many attempts within seconds. This pattern suggests user error (e.g., wrong password) rather than an automated attack.
2. **Account Name:** Some events show "asus" (a valid account), while others show "NULL SID" (a non-existent account). The presence of "NULL SID" is suspicious, as it indicates an attempt with an invalid username.
3. **Logon Type 2:** This is a local logon, meaning the attempt was made physically from the keyboard, not over a network.

Verdict: Suspicious Activity (Not a confirmed attack).

Recommendation: Enable Account Lockout Policy (e.g., lock account after 5 failed attempts) and monitor for increased frequency.

Event ID 4798: User Group Enumeration

Description: This event indicates that a user's local group membership was enumerated (checked).

Observation:

- **Timestamp:** 10/12/2026 1:09:06 PM
- **Account Name:** asus
- **Logon Type:** N/A

Interpretation:

This event shows that the user "asus" checked their local group membership. This is often a normal activity that occurs when software is installed or when a user manually checks their permissions. However, if this activity occurs repeatedly or is performed by an unknown account, it could indicate reconnaissance (information gathering) by an attacker.

Verdict: Normal Activity (Monitor if repeated).

5. Summary of Findings

Event ID	Event Type	Analyzed Count	Verdict
4624	Successful Logon	1	Normal (SYSTEM Service)
4625	Failed Logon	7	Suspicious (Monitor)
4798	Group Enumeration	1	Normal (Monitor if repeated)

(Note: "Analyzed Count" refers to the specific events that were examined in detail for this report.)

6. Conclusion and Learning Outcomes

This project successfully demonstrated how to analyze Windows Security Event Logs to identify potential security threats.

Key Learnings:

1. **Event ID Significance:** Understanding Event IDs (4624, 4625, 4798) is critical for detecting security incidents.
2. **Context Matters:** Not every failed login is a hack. Analysts must consider the Logon Type, Time Frequency, and Source IP before declaring an incident.
3. **Privacy and Ethics:** Masking sensitive data (IP addresses, computer names) is essential when sharing security reports publicly.

7. Disclaimer

All IP addresses, computer names, and usernames in this report have been masked (e.g., XX) to protect the privacy and security of the local environment. Any resemblance to actual public IPs is coincidental.